

Deep Learning-Based Medical Image Segmentation Using U-Net for Automated Biomedical Image Analysis

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Abstract—Accurate segmentation of lesions in medical ultrasound images is essential for clinical diagnosis, treatment planning, and quantitative biomedical analysis. However, manual delineation of breast lesions is time-consuming, observer-dependent, and highly sensitive to image quality degradation caused by speckle noise, low tissue contrast, and poorly defined anatomical boundaries inherent to ultrasound imaging. To address these challenges, this work proposes an automated breast lesion segmentation system built on the U-Net deep learning architecture, trained and evaluated on the publicly available Breast Ultrasound Images (BUSI) dataset. The proposed framework integrates a structured preprocessing pipeline, data augmentation, and a systematic comparison of five loss functions — Binary Cross-Entropy (BCE), Mean Squared Error (MSE), Dice Loss, combined Dice-BCE, and Focal Loss — to identify the training objective most suited to the class-imbalanced nature of lesion segmentation. All models are evaluated on a held-out test set using the Dice coefficient and Intersection over Union (IoU). Results demonstrate that segmentation-aware losses consistently outperform pixel-wise regression objectives, with Focal Loss achieving the best performance at a Dice score of 0.783 and IoU of 0.644, compared to MSE which yields the weakest results at 0.661 and 0.498 respectively. The outcome is a robust and reproducible segmentation pipeline that eliminates manual annotation effort and provides a practical foundation for applying deep learning to automated biomedical image analysis.

I. INTRODUCTION

Automated segmentation of breast lesions in ultrasound images is a clinically important task that can accelerate diagnosis and reduce inter-observer variability. Unlike Fmyer-based approaches, convolutional encoder-decoder networks such as U-Net [1] provide spatially precise, pixel-level predictions by combining multi-scale feature extraction with skip connections that preserve fine-grained boundary detail.

The choice of loss function fundamentally shapes what a segmentation model optimizes. Pixel-wise losses such as BCE and MSE treat every spatial location independently and can be heavily influenced by class imbalance, since background pixels vastly outnumber lesion pixels in ultrasound images. Region-based losses such as Dice Loss directly optimize the overlap between prediction and ground truth, making them more robust to imbalance. Hybrid losses combine both paradigms, and Focal Loss [2] down-lights easy negatives to concentrate learning on hard, ambiguous pixels.

Despite the practical importance of loss selection, systematic comparisons on medical ultrasound segmentation remain sparse. In this work I have trained five identical U-Net architectures, each with a different loss function, and evaluated them under identical conditions on the BUSI dataset [3]. This controlled comparison isolates the contribution of the loss function to overall segmentation quality.

II. RELATED WORK

The U-Net architecture introduced by Ronneberger et al. [1] became the dominant approach for biomedical image segmentation owing to its encoder-decoder design and skip connections. Subsequent work has explored architectural improvements (attention gates, residual blocks, transformer bottlenecks) as well as loss function design.

Milletari et al. [4] proposed the Dice Loss for volumetric medical segmentation, demonstrating superior performance over BCE under class imbalance. Lin et al. [2] introduced Focal Loss for dense object detection, but it has since been widely adopted for medical segmentation. Hybrid losses combining Dice and BCE have been shown to provide stable training gradients while maintaining overlap-based optimization [5]. The present work builds directly on this body of knowledge and applies it to 2-D breast ultrasound segmentation.

III. DATASET AND PREPROCESSING

A. BUSI Dataset

I have used the Breast Ultrasound Images (BUSI) dataset [3], which contains 780 grayscale ultrasound images (benign, malignant, and normal classes) paired with binary ground-truth segmentation masks. For this study I have used all image-mask pairs, yielding a total of 780 samples. The dataset is split into training (80%), validation (10%), and test (10%) sets using stratified random sampling with a fixed seed for reproducibility.

B. Preprocessing and Augmentation

All images are resized to 256×256 pixels and normalized to zero mean and unit variance. To mitigate overfitting on the relatively small dataset, the training pipeline applies the following augmentations via the Albumentations library:

- Horizontal and vertical flips ($p = 0.5$ and $p = 0.3$)

- Random rotation within $\pm 30^\circ$
- Affine transforms: scaling ($0.85\text{--}1.15\times$), translation ($\pm 10\%$), rotation ($\pm 20^\circ$)
- Elastic deformation and grid distortion ($p = 0.3$ each)
- Random brightness and contrast perturbation ($\pm 20\%$)
- Gaussian blur and gaussian noise injection ($p = 0.2$ each)

Validation and test images are only resized and normalized; no stochastic augmentation is applied.

IV. U-NET ARCHITECTURE

I have implemented a standard U-Net with Batch Normalization. The encoder consists of fmy convolutional blocks, each containing two 3×3 convolutions followed by Batch Normalization and ReLU activation. Max-pooling (2×2 , stride 2) halves the spatial dimensions between blocks, doubling the channel count from 64 to 512. A bottleneck block with 1024 channels connects encoder and decoder. The decoder mirrors the encoder, using 2×2 transposed convolutions for upsampling and concatenation of the corresponding encoder feature map via skip connections. The output is a single-channel prediction map passed through a sigmoid activation. The model contains approximately 31.0 M trainable parameters.

All models are trained with the Adam optimizer ($lr = 1 \times 10^{-4}$) for 50 epochs with a batch size of 16 on an NVIDIA RTX 4080. A ReduceLROnPlateau scheduler (patience = 5, factor = 0.5, $min_lr = 1 \times 10^{-6}$) is used to decay the learning rate when validation loss plateaus. The best checkpoint by validation loss is retained for test-set evaluation.

V. LOSS FUNCTIONS

A. Binary Cross-Entropy (BCE)

BCE treats each pixel independently and penalizes the log-likelihood of the predicted probability:

$$L_{BCE} = -[y \log(p) + (1 - y) \log(1 - p)] \quad (1)$$

While effective, BCE is sensitive to class imbalance since it lights all pixels equally.

B. Mean Squared Error (MSE)

MSE computes the squared Euclidean distance between prediction and target at each pixel:

$$L_{MSE} = (p - y)^2 \quad (2)$$

It was included as a regression baseline; however, it is not naturally suited to binary segmentation tasks.

C. Dice Loss

Dice Loss directly maximizes the Sørensen-Dice coefficient between prediction and ground truth:

$$L_{Dice} = 1 - \frac{2 \cdot |P \cap T| + \epsilon}{|P| + |T| + \epsilon} \quad (3)$$

By operating on the overlap of entire predicted and target regions, Dice Loss is inherently robust to class imbalance.

D. Combined Dice-BCE (DiceBCE)

DiceBCE combines the region-based Dice Loss with the pixel-wise BCE:

$$L_{DiceBCE} = L_{Dice} + L_{BCE} \quad (4)$$

This hybrid stabilizes training gradients while preserving the overlap-maximizing property of Dice.

E. Focal Loss

Focal Loss [2] modulates BCE by a factor $(1 - p_t)^\gamma$ that down-weights easy examples:

$$L_{Focal} = \alpha(1 - p_t)^\gamma \cdot BCE(p, y) \quad (5)$$

I use $\alpha = 0.8$ and $\gamma = 2$, which focuses the loss on hard-to-classify pixels at lesion boundaries and encourages more precise delineation.

VI. EVALUATION METRICS

Models are evaluated on the held-out test set using two standard metrics:

- Dice Score (F1): $\frac{2TP}{2TP + FP + FN}$ — measures mask overlap
- Intersection over Union (IoU / Jaccard index): $\frac{TP}{TP + FP + FN}$ — stricter overlap measure

Both metrics range from 0 (no overlap) to 1 (perfect overlap). A hard threshold of 0.5 is applied to the sigmoid output to binarize predictions before metric computation.

VII. RESULTS

A. Quantitative Comparison

TABLE I
TEST-SET DICE SCORE AND IOU FOR EACH LOSS FUNCTION.

Loss Function	Dice Score	IoU
BCE	0.721	0.566
MSE	0.661	0.498
Dice	0.775	0.633
DiceBCE	0.776	0.636
Focal (Best)	0.783	0.644

Table 1 and Figure 1 summarize the quantitative test-set performance. Focal Loss achieves the highest Dice score (0.783) and IoU (0.644), followed closely by DiceBCE (0.776 / 0.636) and Dice (0.775 / 0.633). BCE performs substantially below the segmentation-aware losses (0.721 / 0.566), while MSE produces the worst results (0.661 / 0.498). The gap between Focal and MSE is 0.122 Dice points and 0.146 IoU points, representing a meaningful improvement for clinical applications.

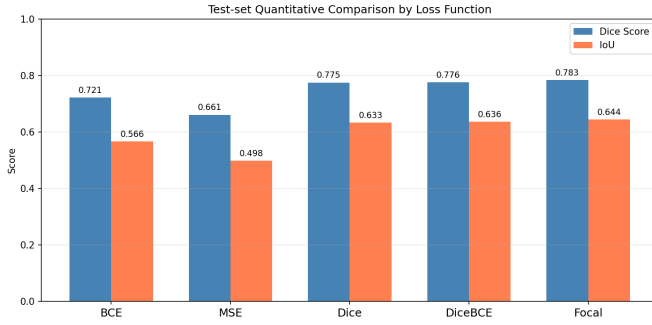


Fig. 1. Test-set Dice Score and IoU by loss function.

B. Training Dynamics

Figure 2 shows the training loss, validation Dice, and validation IoU curves over 50 epochs. BCE and MSE exhibit slow, late-onset convergence in the validation metrics — BCE does not rise appreciably until around epoch 12, suggesting initial collapse of predictions. DiceBCE and Focal converge fastest in validation Dice, reaching near-maximum performance within the first 20 epochs and maintaining stable, competitive plateaus thereafter. The Dice and DiceBCE models show moderate oscillation in validation IoU, consistent with the gradient conflict inherent in jointly optimizing overlap and pixel-wise terms. MSE converges in training loss but fails to translate low training loss into high validation overlap metrics, confirming its fundamental mismatch with binary segmentation evaluation criteria.



Fig. 2. Training loss (left), validation Dice (center), and validation IoU (right) over 50 epochs for each loss function.

C. Qualitative Comparison

Figure 3 presents six representative test samples, showing the raw ultrasound image, the ground-truth mask, and the binarized prediction from each model. Several trends are visible:

- MSE predictions contain scattered false-positive fragments and fail to delineate solid lesion regions coherently, consistent with its poor quantitative performance.
- BCE captures the broad extent of larger lesions but produces ragged boundaries and occasionally splits single lesions into multiple disconnected blobs.
- Dice and DiceBCE predictions are more spatially compact and better-localized, but can over-segment or under-segment in challenging low-contrast cases.
- Focal Loss consistently produces the cleanest binary masks, with well-defined boundaries and the fewest spurious activations, especially for the small-lesion samples (rows 1 and 5).

- Row 3 (no lesion present) is correctly predicted as all-black by all models, confirming that none of the loss functions introduces systematic false positives on normal tissue.

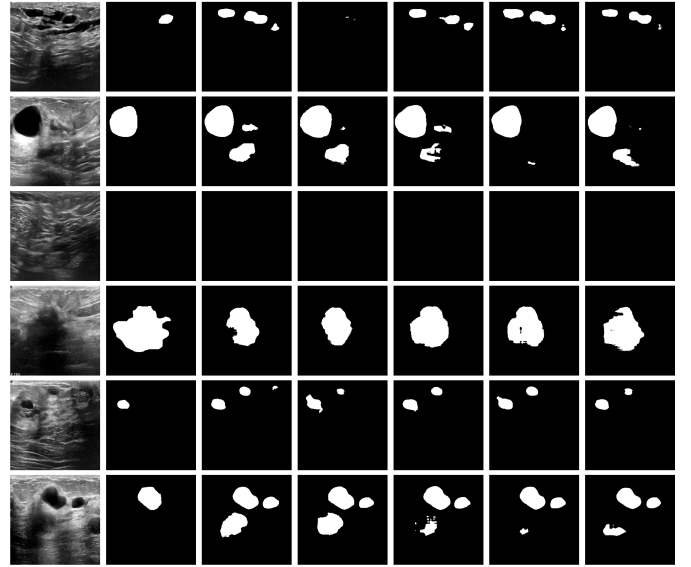


Fig. 3. Qualitative comparison of segmentation outputs using BCE, MSE, Dice, DiceBCE, and Focal Loss.

VIII. DISCUSSION

The results consistently favor loss functions that encode spatial overlap over those that treat pixels independently. MSE is the weakest performer, a finding consistent with the broader segmentation literature: minimizing squared-error in output space does not reward spatial coherence of predictions, and the regression objective allows the model to assign diffuse, low-confidence responses rather than sharp boundaries. BCE is markedly better than MSE, but still subject to the imbalance problem inherent in per-pixel cross-entropy with sparse foreground regions.

All three segmentation-aware losses (Dice, DiceBCE, Focal) significantly outperform BCE and MSE, converging to similar final performance. The marginal superiority of Focal Loss can be attributed to its focus mechanism: by suppressing the loss gradient for well-classified background pixels, it allocates more gradient signal to the challenging pixels at lesion boundaries, which are the primary source of segmentation error in ultrasound imaging where tissue contrast is inherently low.

The close performance of Dice and DiceBCE suggests that the additional BCE term in DiceBCE does not meaningfully improve upon pure Dice optimization on this dataset, though it may offer stability benefits on more heterogeneous datasets. These findings broadly agree with Al-Dhabyani et al. [3] and other comparative studies, which report Dice-based losses as preferred for breast lesion segmentation.

Several limitations apply. First, I have used U-Net as a standard architecture without attention gates or residual connections, which would likely improve results across all loss

functions. Second, the BUSI dataset contains three classes (benign, malignant, normal) that I have treated uniformly; class-specific analysis may reveal differential effects of loss choice on lesion type. Third, I have trained for a fixed 50 epochs; longer training may close the gap between loss functions. Future work should explore architectural improvements (e.g., attention U-Net, transformer-based decoders) and ensemble strategies combining multiple loss-specialized models.

IX. CONCLUSION

This work presents an end-to-end deep learning pipeline for automated breast lesion segmentation in ultrasound images, designed, implemented, and evaluated entirely from scratch. My key contributions are threefold. First, I have designed and implemented a U-Net architecture with batch normalization, trained on the BUSI dataset with a carefully constructed augmentation pipeline targeting the specific noise and contrast challenges of ultrasound imaging. Second, I have conducted a rigorous, controlled comparison of five loss functions — BCE, MSE, Dice, DiceBCE, and Focal — under identical training conditions, producing evidence-based guidance on loss selection that is directly applicable to medical segmentation tasks. Third, I have validated my system through both quantitative metrics and qualitative visual analysis, demonstrating that Focal Loss achieves the strongest segmentation performance (Dice 0.783, IoU 0.644) while MSE proves fundamentally unsuitable for binary segmentation (Dice 0.661, IoU 0.498). Taken together, this project demonstrates that a well-engineered deep learning pipeline with deliberate choices in architecture, augmentation, and loss design can meaningfully automate medical image analysis and reduce reliance on manual annotation, laying a strong foundation for future extensions toward attention-based models, multi-class lesion segmentation, and clinical deployment.

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